
Microeconomics I

Class 3 — Utility representation

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Short quiz

8 minutes · pens down at 10:08

Circle the letter of your answer in the block at the top of the sheet.

Changed your mind? Cross out that circle and circle the new letter.

Write your name on the block, tear the strip off along the dashed line, hand in *only the strip*.

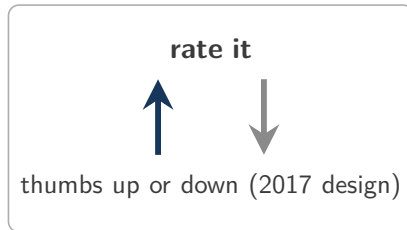
Plan for today

10:00	Short quiz
10:08	Practicalities; five stars or a thumb
10:15	Utility functions: what the numbers mean
10:28	Continuity, existence, the lexicographic case
10:50	Break
11:00	Quasiconcavity and concavity
11:23	Common utility functions
11:40	Summary; problem set 1

Practicalities

- Problem set 1 today, on paper: one copy per group, your group's own numbers
- Due in class Wed 14 Oct, one paper copy per group, all names on it
- Office hours: Wednesdays 12:00–13:00, after class
up to 20 minutes per student; those who book by e-mail have priority
- Slides and reading: pkocourek.com/teaching/micro-1 · pkocourek@ub.edu

Five stars or a thumb?



Which of the two is a utility function?

Netflix: stars until April 2017, thumbs since (a double thumb-up added 2022). Evidence, not a recommendation.

Thumbs instead of stars: three times as many answers

thumbs instead of stars

+200 %

ratings given, in tests with
hundreds of thousands of members

April 2017

stars gone

the star was a guess for you,
often read as an average;
now a personal “% match”

Stars invite **arithmetic**; a thumb does not. Does the theory need it? Back to it on slide 10.

Netflix, “Goodbye Stars, Hello Thumbs”, 5 April 2017; announced 16 March 2017 (Variety, TechCrunch).

Definition: a utility function represents $\succsim$

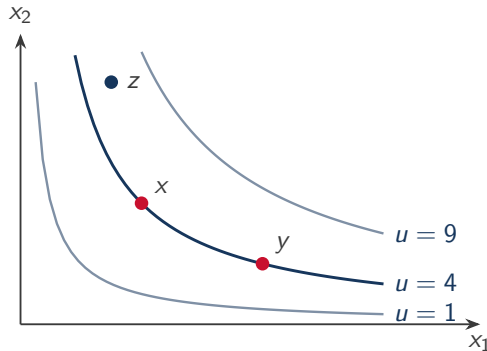
$u : X \rightarrow \mathbb{R}$ **represents** $\succsim$ if, for all $x, y \in X$,

$$x \succsim y \iff u(x) \geq u(y)$$

Represented $\Rightarrow \succsim$ is rational

complete and transitive, because $\geq$ on $\mathbb{R}$ is

MWG Definition 1.B.2, Proposition 1.B.2.



$u(x) = x_1x_2$: $u(x) = u(y) = 4$, so $x \sim y$; $u(z) = 6$, so $z \succ x$.

From class 2: four of the five were already “ranked by a number”

	$x \succsim y \iff$	$u(x) =$
perfect substitutes	$x_1 + x_2 \geq y_1 + y_2$	$x_1 + x_2$
perfect complements	$\min\{x_1, x_2\} \geq \min\{y_1, y_2\}$	$\min\{x_1, x_2\}$
Cobb–Douglas type	$x_1 x_2 \geq y_1 y_2$	$x_1 x_2$
quasilinear	$x_1 + \sqrt{x_2} \geq y_1 + \sqrt{y_2}$	$x_1 + \sqrt{x_2}$
lexicographic	$x_1 > y_1$, or $x_1 = y_1$ and $x_2 \geq y_2$	?

Three questions for today: is the number **unique**? When does one **exist**? What do its **properties** say about $\succsim$?

In pairs: same preference as $u(x) = x_1x_2$?

$v(x) =$ same $\succsim$?

$$\sqrt{x_1x_2}$$
$$2 - \frac{1}{1 + x_1x_2}$$
$$x_1 + x_2$$
$$x_1x_2^2$$

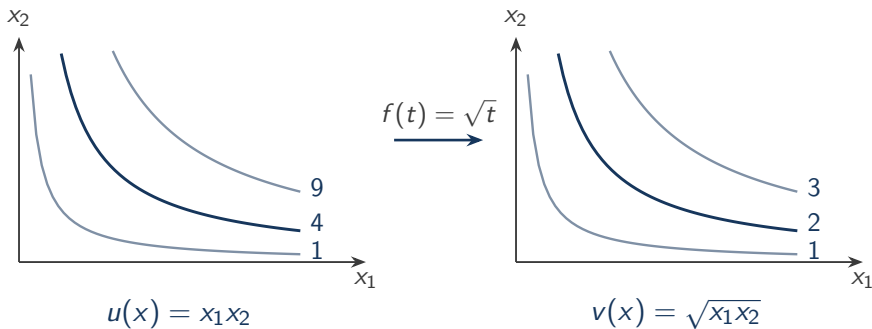
Four minutes. Same preference means: the same ranking of every pair of bundles.

In pairs: same preference as $u(x) = x_1x_2$?

$v(x) =$	same $\succsim$?	
$\sqrt{x_1x_2}$	✓	$v = f(u)$ with $f(t) = \sqrt{t}$, strictly increasing
$2 - \frac{1}{1 + x_1x_2}$	✓	$f(t) = 2 - \frac{1}{1+t}$, strictly increasing; all values below 2
$x_1 + x_2$	✗	(1, 4) and (2, 2): u says $\sim$, v says $5 > 4$
$x_1x_2^2$	✗	(1, 4) and (2, 2): u says $\sim$, v says $16 > 8$

Four minutes. Same preference means: the same ranking of every pair of bundles.
Two keep the ranking, one of them bounded. Two disagree already at (1, 4) vs (2, 2).

The numbers are labels: any strictly increasing relabelling will do



u represents $\succsim$ and $f : \mathbb{R} \rightarrow \mathbb{R}$ strictly increasing $\Rightarrow v = f \circ u$ represents $\succsim$

Utility is **ordinal**: same map, different numbers on the curves. Differences of u mean nothing.

MWG 3.C, after Proposition 3.C.1; Section 1.B. Proof: $v(x) \geq v(y) \iff u(x) \geq u(y) \iff x \succsim y$.

Back to the stars

stars invite arithmetic

★★★★★ + ★★★ = 3?

averages and gaps: a **cardinal** use

a thumb

liked it, or not

nothing to add up: **ordinal** only

The theory needs only the ranking. Nothing in $\succsim$ says how much better.

Neither screen is a utility function: both coarsen the ranking, like $f(t) = \min\{t, 4\}$. Netflix's members read a personal prediction as an average: arithmetic the theory never does.

Continuity and existence

MWG 3.C

Definition: continuous preferences

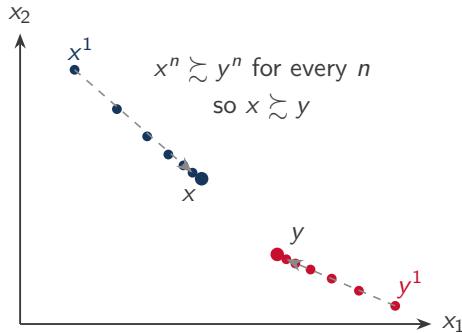
$\succsim$ on X is **continuous** if it is preserved under limits:

$$x^n \succsim y^n \text{ for all } n,$$

$$x = \lim x^n, \quad y = \lim y^n$$

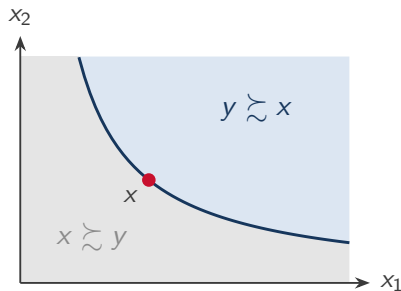
$$\Rightarrow x \succsim y$$

MWG Definition 3.C.1.



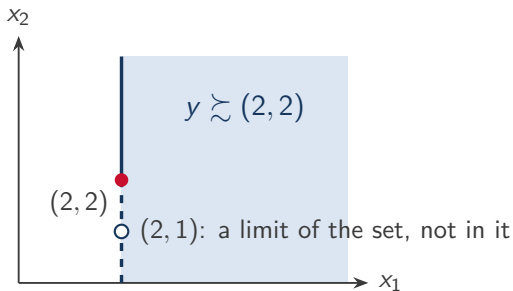
No sudden reversal at the limit: what holds along the whole sequence holds at the end.

Equivalently: upper and lower contour sets are closed



continuous

the curve belongs to both sets: closed



lexicographic

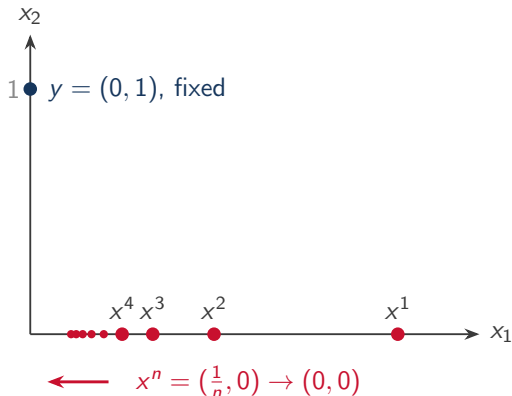
the dashed edge is missing: not closed

Closed: limits of points of the set stay in the set.

Right: $(2 + \frac{1}{n}, 1) \succsim (2, 2)$ for every n , but the limit $(2, 1)$ is not in the set.

MWG 3.C, text after Definition 3.C.1 (one direction; the converse is Exercise 3.C.3).

Lexicographic preferences reverse at the limit



$x^n \succ y$ for every n

$\frac{1}{n} > 0$: more of good 1 decides

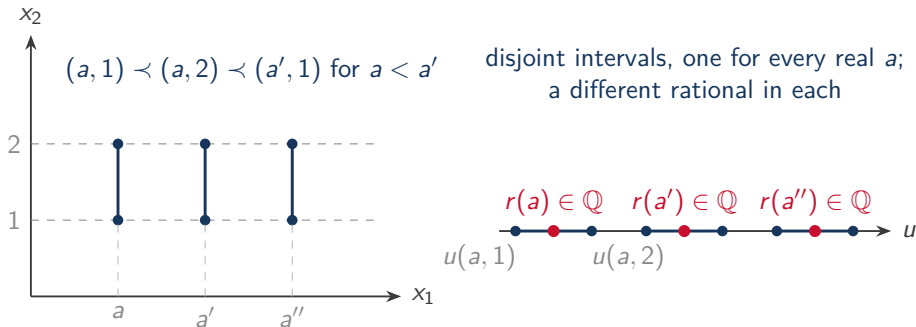
$\lim x^n = (0, 0) \prec (0, 1) = y$

tie in good 1: now good 2 decides

Not continuous.

The ranking flips exactly where the first coordinates become equal. Example 3.C.1, continued.

Lexicographic preferences have no utility function at all



A one-to-one map from the reals into the rationals: uncountable into countable. Impossible.

MWG Example 3.C.1, small print on p. 46. This rules out every u , continuous or not.

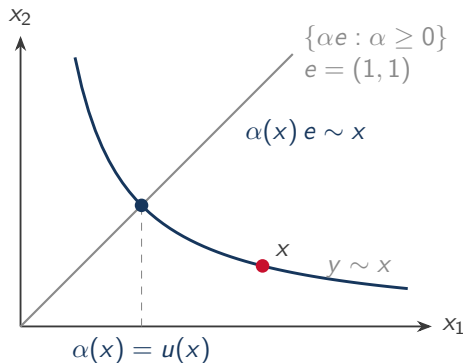
Continuity is enough: a continuous utility function exists

$\succsim$ rational and continuous on X
 $\Rightarrow$ some continuous $u : X \rightarrow \mathbb{R}$
represents $\succsim$

Proof idea, $\succsim$ monotone:

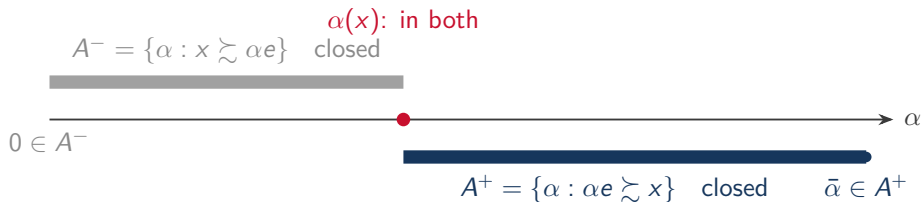
$$u(x) = \alpha(x) \text{ with } \alpha(x) e \sim x$$

MWG Proposition 3.C.1, Figure 3.C.1 (monotone case).



Each curve is labelled by where it crosses the diagonal; further out crosses further along.

Why every curve crosses the diagonal, once



closed (continuous), nonempty (monotone), cover $\mathbb{R}_+$ (complete) $\Rightarrow$ they overlap

monotone: $\alpha' > \alpha \Rightarrow \alpha' e \succ \alpha e$, so they overlap in **one** point

$$u(x) \geq u(y) \iff \alpha(x)e \succsim \alpha(y)e \iff x \succsim y$$

MWG, proof of Prop. 3.C.1, pp. 47–48. $\mathbb{R}_+$ is connected: two nonempty closed sets covering it meet.

In pairs: continuous? represented by some u ?

$$x \succsim y \iff$$

continuous some u

$$x_1 x_2 \geq y_1 y_2$$

$$x_1 > y_1, \text{ or } x_1 = y_1 \text{ and } x_2 \geq y_2$$

$$\lfloor x_1 \rfloor \geq \lfloor y_1 \rfloor \quad (\text{whole units of good 1 only})$$

$$\|x - (5, 5)\| \leq \|y - (5, 5)\|$$

Six minutes. For “not continuous”, give a sequence that reverses at the limit.

In pairs: continuous? represented by some u ?

$x \succsim y \iff$	continuous	some u	
$x_1 x_2 \geq y_1 y_2$	✓	✓	$u = x_1 x_2$
$x_1 > y_1$, or $x_1 = y_1$ and $x_2 \geq y_2$	✗	✗	slides 14–15
$\lfloor x_1 \rfloor \geq \lfloor y_1 \rfloor$ (whole units of good 1 only)	✗	✓	$u = \lfloor x_1 \rfloor$
$\ x - (5, 5)\ \leq \ y - (5, 5)\ $	✓	✓	$u = -\ x - (5, 5)\ $

Six minutes. For “not continuous”, give a sequence that reverses at the limit.

Row 3: $(2 - \frac{1}{n}, 0) \succsim (1, 0)$ for all n , the limit $(2, 0)$ is not; continuity is **sufficient**, not necessary.

Row 4: closed discs; not monotone, and it does not matter.

Break

Back at 11:00

Quasiconcavity and concavity

MWG 3.C

Definition: quasiconcave utility

u is **quasiconcave** if every set $\{y \in X : u(y) \geq u(x)\}$ is convex, that is,

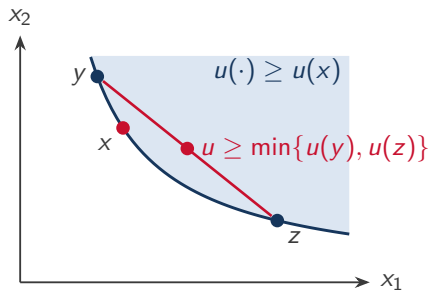
$$u(\alpha x + (1 - \alpha)y) \geq \min\{u(x), u(y)\}$$

for all $x, y \in X$ and $\alpha \in [0, 1]$; **strictly**: $>$ when $x \neq y$, $\alpha \in (0, 1)$

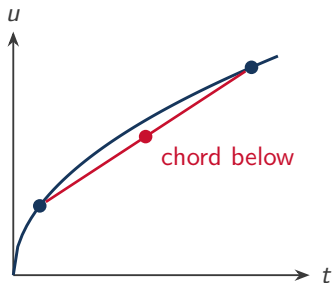
$\succsim$ convex $\iff u$ quasiconcave
strictly convex $\iff$ strictly quasiconcave

MWG 3.C, pp. 49–50.

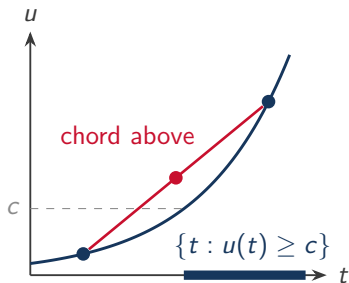
Class 2's picture: the upper contour sets of $\succsim$ are the sets $\{u \geq u(x)\}$.



Concave is stronger, and it is about the numbers



concave: $u(t) = \sqrt{t}$
and quasiconcave



quasiconcave, not concave: $u(t) = e^{0.6t}$
increasing, so every level set is a half-line: convex

concave: $u(\alpha x + (1 - \alpha)y) \geq \alpha u(x) + (1 - \alpha) u(y) \Rightarrow$ quasiconcave, not $\Leftarrow$
MWG 3.C, p. 50: some convex $\succsim$ have no concave representation at all.

What survives a strictly increasing relabelling $v = f \circ u$

	survives	
represents the same $\succsim$	✓	that is what f increasing means
increasing ($x \gg y \Rightarrow v(x) > v(y)$)	✓	monotone $\succsim$
quasiconcave	✓	same level sets, same $\succsim$: convex $\succsim$
the MRS at every bundle	✓	$f'(u) u_1 / f'(u) u_2 = u_1 / u_2$; f differentiable, $f' > 0$
concave	✗	$x_1 x_2$ vs $\sqrt{x_1 x_2}$: next slide
marginal utility $\partial u / \partial x_1$	✗	rescaled by f'
continuous	✗	a jump in f makes v jump

Top: **ordinal**, facts about $\succsim$. Bottom: **cardinal**, facts about one convenient u .

MWG 3.C, p. 50 and footnote 6.

In pairs: quasiconcave? concave?

$u(x) =$	quasiconcave	concave
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$x_1 x_2$

$\sqrt{x_1 x_2}$

$x_1 + x_2$

$(x_1 + x_2)^2$

Five minutes. Test concavity at $(0, 0)$, $(2, 2)$ and their midpoint.

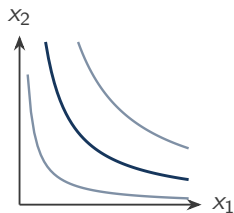
In pairs: quasiconcave? concave?

$u(x) =$	quasiconcave	concave	
$x_1 x_2$	✓	✗	$u(1, 1) = 1 < 2 = \frac{1}{2}u(0, 0) + \frac{1}{2}u(2, 2)$
$\sqrt{x_1 x_2}$	✓	✓	same $\succsim$ as row 1; the geometric mean is concave
$x_1 + x_2$	✓	✓	linear: both, neither strictly
$(x_1 + x_2)^2$	✓	✗	same $\succsim$ as row 3; $u(1, 1) = 4 < 8$

Five minutes. Test concavity at $(0, 0)$, $(2, 2)$ and their midpoint.

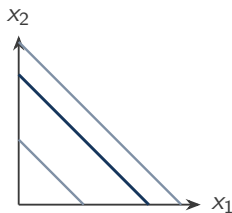
Rows 1–2 and 3–4 are the same preferences: quasiconcavity agrees, concavity does not.

Four utility functions you will use all month



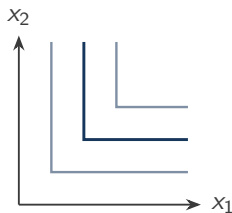
Cobb–Douglas

$$x_1^\alpha x_2^\beta$$
$$\text{MRS} = \frac{\alpha x_2}{\beta x_1}$$



perfect substitutes

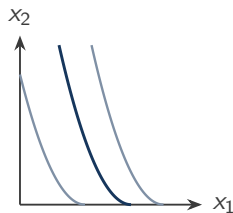
$$ax_1 + bx_2$$
$$\text{MRS} = \frac{a}{b}$$



Leontief

$$\min\{ax_1, bx_2\}$$

no MRS at the kink

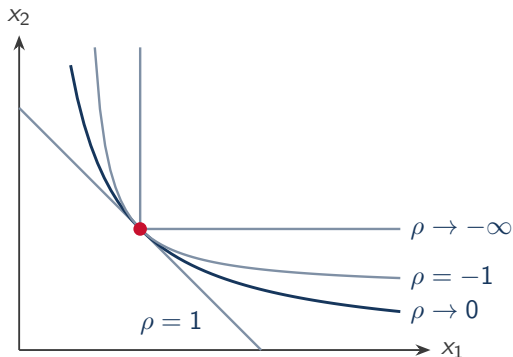


quasilinear

$$x_1 + \varphi(x_2)$$
$$\text{MRS} = \frac{1}{\varphi'(x_2)}$$

$\alpha, \beta, a, b > 0$; φ increasing and concave. $\text{MRS}_{12} = u_1/u_2$, class 2. Leontief kinks: MWG Figure 3.C.3.

One formula holds the whole family: CES



$$u(x) = \left(\frac{1}{2}x_1^\rho + \frac{1}{2}x_2^\rho \right)^{1/\rho}$$

$\rho \leq 1$, $\rho \neq 0$; for $\rho < 0$ take $x \gg 0$

$\rho = 1$: perfect substitutes $\frac{1}{2}(x_1 + x_2)$

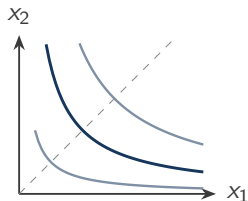
$\rho \rightarrow 0$: Cobb–Douglas $\sqrt{x_1 x_2}$

$\rho \rightarrow -\infty$: Leontief $\min\{x_1, x_2\}$

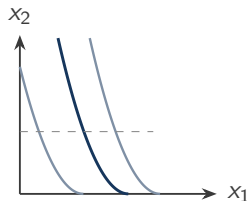
All four curves are $u = 2$. Lower ρ : a sharper bend, the goods substitute less easily.

Elasticity of substitution $\sigma = 1/(1 - \rho)$; Muñoz-García §1.8.4 (with weights a, b). Returns with demand.

Class 2's two families, in utility form



homothetic



quasilinear in good 1

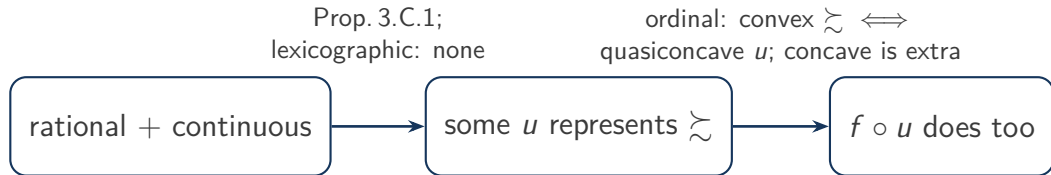
continuous $\succsim$: **homothetic** $\iff$ some u with $u(\alpha x) = \alpha u(x)$ for $\alpha > 0$
 $\sqrt{x_1 x_2}, \quad x_1 + x_2, \quad \min\{x_1, x_2\}$

continuous $\succsim$: **quasilinear** in good 1 $\iff$ some u of the form $x_1 + \varphi(x_2)$
 $x_1 + \sqrt{x_2}, \quad x_1 + \ln(1 + x_2)$

“Some u ”: a cardinal choice. $x_1 x_2$ is homothetic too, and not homogeneous of degree one.

MWG Exercise 3.C.5, stated on p. 50; quasilinear on $X = \mathbb{R} \times \mathbb{R}_+$. Both return with demand next week.

Today in one line



Wednesday: budget set, utility maximisation, Walrasian demand.

Problem set 1: one copy per group, due Wed 14 Oct.

Read MWG 2.D and 3.D (or Muñoz-García ch. 2) for Wednesday 7 October.